

NPSC CTSE 2026 — Official Pattern & Syllabus (Ground Truth)

Everything in this file is transcribed verbatim from official NPSC sources.

Source PDFs are in [_raw/syllabus/](#). Do not study from anything that contradicts this file.

- Pattern: <https://npsc.nagaland.gov.in/storage/cms/media/2025/174306246210129.pdf>
- CS (Degree): <https://npsc.nagaland.gov.in/storage/cms/media/2025/174158375357771.pdf>
- CS (Diploma): <https://npsc.nagaland.gov.in/storage/cms/media/2025/174066932242843.pdf>
- Computer Forensic: <https://npsc.nagaland.gov.in/storage/cms/media/2026/177744361268586.pdf>

1. PATTERN OF EXAMINATION (official, verbatim)

The Written Examination will consist of **4 (four) papers**:

#	Paper	Marks	Duration
1.1	General English & General Knowledge (Combined)	100	3 Hours
1.2.1	Technical Subject — MCQ (100 Questions × 2 marks)	200	2 Hours
1.2.2	Technical Subject — Descriptive Paper-I	100	3 Hours
1.2.3	Technical Subject — Descriptive Paper-II	100	3 Hours
	Grand Total (per elective)	500	

⚠ **MCQ has negative marking of 'one third' ($-1/3 \times 2 = -0.667$) for each wrong answer.**

Components of the Combined General English & GK paper

2.1 General English — 50 Marks

Component	Marks
2.1.1 Essay	1 × 20
2.1.2 Précis writing	1 × 10
2.1.3 Amplification	1 × 10
2.1.4 Construction of sentence & Grammar	10

2.2 General Knowledge — 50 Marks

"General knowledge comprising of current events, general science, mental ability, Indian history, Indian polity, Indian economy, Geography with combination of objective type and short descriptive answers — 50 Marks"

Total (2.1 + 2.2) = 100 Marks

Observed descriptive paper structure (from CTSE 2025 CS Degree Paper-I)

Maximum marks: 100 Duration: 180 minutes
Section A: answer any 8 of Q1-Q10 (8 × 5 = 40 marks)
Section B: answer any 3 of Q11-Q15 (3 × 10 = 30 marks)
Section C: answer any 2 of Q16-Q19 (2 × 15 = 30 marks)

This is the single most important structural fact for the descriptive papers.

You get *choice* — 10 questions offered in Section A, you answer 8. You can skip your two weakest topics in every section.

2. ELECTIVE 1 — COMPUTER FORENSIC

Technical Paper-I (TP-I)

1. **INTRODUCTION TO FORENSIC SCIENCE** — Definition, History, Development and Scope of Forensic Science in India. Basic Principles of Forensic Science and its Significance, Organization and Functioning of State and Central Forensic Science Laboratories, Ethics in Forensic Science.
2. **PHYSICAL EVIDENCE** — Definition, Types, Class, and Individual Characteristics, Different Searching methods for locating physical evidences at scene of crime, chain of custody.
3. **SCENE OF CRIME** — Meaning, Types, Protection of Scene of Crime, Crime Scene Documentation — Note Taking, Videography, Photography and Sketching Methods, 3-D Scanning Technique. Searching Technique of Crime Scene, Processing of Physical Evidence Discovering, Recognizing and Examination of Physical Evidence, Collection, Safety measure for Evidence Collection, Preservation, Packaging, Sealing, Labeling and Forwarding of Physical Evidence.
4. **INTRODUCTION TO COMPUTER HARDWARE** — Various components of a Computer, Motherboard, Processor, Memory, Storage Devices and Networking components. Understanding Computer Operating Systems (OS), Booting process of computers. Introduction to file systems and types of file system.
5. **WINDOWS SYSTEMS ARTIFACTS** — File Systems, Registry, Event logs, Shortcut files, Executables. Alternate Data Streams (ADS), Hidden files, Slack Space. **Linux System and Artifacts:** Linux file system — Ownership and permissions, hidden files, User Accounts and Logs. **Mac OS X systems and Artifacts:** System startup and Services, Network configuration, hidden directories, system logs and user artifacts.
6. **LOGIC, FORMULA, UNIT, ETC** — Propositional (Boolean) Logic, Predicate Logic, Well-Formed Formulae (WFF), Satisfiability and Tautology. Digital Computers, Logic Gates, Boolean Algebra, map Simplifications, Combinational Circuits, Flip-Flops, Sequential Circuits, Integrated Circuits, Decoders, Multiplexers, Registers and Counters, Memory Unit, Representation of Integers: Octal, Hex, Decimal, and Binary. 1's complement and 2's complement arithmetic. Floating point representation.
7. **COMPUTER NETWORKS** — Network fundamentals: LAN, MAN, WAN, Wireless Networks, Inter Networks. Reference Models: The OSI model, TCP/IP model, IP address and Mac address. Data Communication: Channel capacity, Transmission media — twisted pair, coaxial cables, fibre-optic cables, wireless transmission — radio, microwave, infrared waves, Lightwave transmission, Telephone — local loop, trunks, multiplexing, switching, narrowband ISDN, broadband ISDN, ATM, High-speed LANS, Cellular and Radio. Communication Internetworking: Switch/Hub, Bridge, Router, Gateways, Concatenated virtual circuits, Tunnelling, Fragmentation, Firewalls, Routing, Virtual circuits and datagrams, Routing algorithms, Network Security, Cryptography — public key, secret key, Domain Name System (DNS), Electronic Mail and World Wide Web (WWW), the DNS, Resource Records, Name Servers, E-Mail Architecture.
8. **OPERATING SYSTEMS** — Main functions of Operating Systems. Multi Programming, multiprocessing, and multitasking. Memory Management: Virtual memory, paging, fragmentation. Concurrent Processing: Mutual exclusion, Critical regions, lock and unlock. Scheduling: CPU scheduling, I/O scheduling, Resource scheduling, Deadlock and scheduling algorithms. Disk Management — formatting, boot block, disk free space management techniques, concept of RAID. Protection and Security: Concepts of domain, access control, Security of systems — concepts, threats — Trojan horse, virus, worms, etc, User authentication. **UNIX:** The Unix System: File system, process management, shell variables, command line programming. Filters and Commands. **Windows Operating Systems:** Design Principles, System Components, Terminal Services and Fast User Switching; File System, Networking.
9. **COMPUTER ORGANIZATION AND ARCHITECTURE** — Introduction to basic structures and operational concepts, Instruction formats, Instruction execution, sequencing, Addressing modes, Stacks, Queues,

Subroutines Concepts, Fetching and Storing word from/in main memory, Register transfers, Operations, execution of a complete instruction, Basic concepts, RAM, ROM — different types, Characteristics, Cache memories, Performance (memory interleaving, hit rate, etc.), Memory hierarchy — virtual memory — address translation, secondary memories, input/output organization, memory mapped, standard (isolated) and linear selection techniques of I/O addressing. Data transfer through programmed I/O, interrupt and DMA I/O processors. Data transfer over synchronous and asynchronous buses; discussions on some standard interface buses.

Technical Paper-II (TP-II)

1. **CYBER CRIME** — Form of Cyber Crime, Internal and External Attacks, Crimes related to Social Media, ATM and Banking Frauds. Data Privacy issues, Packet sniffing, Spoofing, web Security. First responder — role and toolkit. Procedure for search and seizure of digital evidences. Search and Seizure of Volatile and Non-volatile digital evidence. Imaging and Hashing digital evidence. Analyzing and Recovery of deleted, hidden and altered files.
2. **WEB BROWSERS** — Cookies, Favourites or Bookmarks, Cache, Session Data and Plugins. Email: Types of Email and Protocols. Analysing the header details and tracking the email, spoofed mails. Virtual machine and Cloud technology forensics.
3. **SOFTWARE ENGINEERING** — System Development Life Cycle (SDLC): Steps, Waterfall model, Prototypes, Spiral model. Software Metrics: Software Project Management. Software Design: System design, detailed design, function oriented design, object oriented design, user interface design, design level metrics. Coding and Testing: Testing level metrics. Software quality and reliability, clean room approach, software reengineering, reverse engineering.
4. **PROGRAMMING IN C** — Elements of C: Tokens, identifiers, data types in C, control structures in C, sequence, selection and iteration(s). Structured data types in C — arrays, struct, union, string, and pointers. **Object Oriented Programming Concepts**: Class, object, instantiation, inheritance, polymorphism and overloading. **C++ Programming**: Elements of C++ tokens, identifiers, variables and constants, datatypes, operators, control statements, Functions parameter passing, class and objects, constructors and destructor, overloading, inheritance, templates, exception handling. **Web programming**: HTML, DHTML, XML, Scripting, Java, Servlets, Applets.
5. **DATABASE AND SQL** — E-R diagrams and their transformation to relational design, normalization. SQL: Data Definition and Data Types; Constraints, Queries, Insert, Delete, and Update Statements; Views, Stored Procedures and Functions; Database Triggers, SQL Injection, Transaction Processing, concurrency Control Techniques, Database Recovery Techniques, Object and Object-relational Database; Database Security and Authorization, data dictionary.
6. **DATA AND FILE STRUCTURES** — Data, Information, Definition of data structure. Arrays, stacks, queues, linked lists, trees, graphs, priority queues and heaps. File structures: Fields, records and files. Sequential, direct, index-sequential and relative files. Hashing, inverted lists and multi-lists. B-trees and B+ trees.
7. **MOBILE COMPUTING** — Mobile connectivity — cells, Framework, wireless delivery technology and switching methods, mobile information access devices, mobile data internetworking standards, cellular data communication protocols, mobile computing applications. Mobile databases — protocols, scope, tools and technology, M-business.
8. **DIGITAL EVIDENCES** — Digital Evidence storage devices, Order of volatility of digital evidence, cybercrime and types, Cyber Forensics, basic rules for cyber forensics, collection and preservation of digital evidence, data acquisition methods, Forensic Imaging and cloning, data validation, Hash algorithms and hash value, various file extensions and file signatures. Windows Registry overview, data recovery techniques. Windows Disks, files and partitions, Email tracing techniques, social media analysis, Steganography and analysis methods.

3. ELECTIVE 2 — COMPUTER SCIENCE & ENGINEERING (DEGREE)

Paper-I

#	Subject	Description
1	Basics and Programming	Basic units of computer systems, algorithms, flowcharts; Programming in C/C++: Data types, declarations and expressions, functions, pointers, arrays.
2	Data Structure & Algorithm	Primitive data types, array, stack, queue, link list, trees, sorting and searching techniques, symbol tables, hashing etc. Analysing Algorithms, the big-Oh notation, divide and conquer & greedy method, Dynamic programming, Search and traversal techniques, NP completeness.
3	Logic Design	Number system, Binary arithmetic, Boolean algebra and logic functions, Minimization, Flip-flops, design of combinational and Sequential circuits, Registers, Counters, decoders, Encoders, Adders circuits and Code converters.
4	Computer Organization & Architecture	Software and Hardware organization of computers, addressing modes, CPU design, Memory organization, I/O organization, DMA Data transfers.
5	System Software's & Compilers	Assemblers, macros, loaders, linkers and editors. Lexical analysis, Syntax analysis — different parsing techniques, semantic analysis, Error detection, Optimization and code generation.
6	Mobile Application & Services	Factors in Developing Mobile Applications, Frameworks and Tools in Mobile Applications, Text-to-Speech Techniques, Designing the Right UI, Multichannel and Multimodal UIs, Storing and Retrieving Data, Synchronization and Replication of Mobile Data, Android Storing and Retrieving Data, Putting It All Together: Packaging and Deploying.

Paper-II

#	Subject	Description
1	Operating Systems	Introduction to Operating system, Evolution, Generations, Components & Services, Process Management, Scheduling algorithms, Memory management, I/O management, File system Management, Security & Protection.
2	Software Engineering	Concept of systems, Software development process modules, software project planning and management; cost estimation methods, scheduling requirement analysis, software testing strategies, Quality Assurance.
3	Data Based Management System	Data Abstraction, Data models, Data independence, DDL, Attributes, Keys, Query processing, Structure of relational Databases, Example of SQL; Distributed Databases, Security and integrity, Network Model.
4	Data Communication & Computer Network	Basic Network Concepts, Network Topologies, Networking devices, Transmission media, OSI Model, TCP/IP fundamentals, Knowledge of Internet, Internet routing, IP address.
5	Web Technologies	The Phases of web site development: Implementation, Maintenance, Testing. Basic HTML Concepts, HTML, HEAD, TITLE, BODY, Paragraphs, Lists, Formatted and Unformatted Text, Hyperlink, Font (Size, Colour), image. PHP: Server-side web scripting, Installing PHP, Adding PHP to HTML, Syntax and Variables, Passing information between pages, Basic PHP error/problems.
6	AI — Artificial Intelligence	Knowledge and Reasoning: Building a Knowledge Base. Learning from Observations: Inductive Learning, Explanation Based Learning. Pattern Recognition: Structured Description, Symbolic Description, Object Identification, Speech Recognition. Programming Language: LISP, PROLOG.
7	Machine Learning	Knowledge-level vs symbol-level learning. Supervised Learning: Distance-based methods, Nearest-Neighbours. Unsupervised Learning Clustering: K-means/Kernel K-means. Statistical Learning Theory: Ensemble Methods (Boosting, Bagging, Random Forests).

4. ELECTIVE 3 — COMPUTER SCIENCE / COMPUTER ENGINEERING (DIPLOMA)

Paper-I

- 1. INTRODUCTION TO COMPUTERS** — 1.1 Generations of Computers (First → Fourth). 1.2 Classification: Super-computers, Mainframe, Mini, Workstations, Micro. 1.3 Data representation and Number systems: Bit, Byte, Nibble, Word; BCD; ASCII; Binary/Octal/Decimal/Hexadecimal; conversions; binary addition, subtraction, 1's and 2's complement. 1.4 Basic organization: Functional Block diagram, Input unit, Output unit, CPU (Main memory, ALU, CU), Mass Storage unit.
- 2. INPUT AND OUTPUT DEVICES** — Input: Keyboard, Mouse (scroll, optical), Scanners (OCR, bar-code reader). Output: Monitors/VDU (CRT, LCD), Printers (Dot Matrix, Laser, Inkjet), Plotters.
- 3. STORAGE DEVICES** — Non-volatile: ROM, PROM, EPROM. Volatile: Registers, Cache, Main memory RAM. Comparative study of cache/primary/secondary w.r.t. capacity and speed. SDRAM vs DDR RAM. Magnetic storage: Floppy, Hard disk (components, organization of data, drive interface). Optical: CD/DVD read/write mechanism.
- 4. COMPONENTS OF MOTHERBOARD** — Memory controller hub (System bus, Processor socket, DIMM slot, AGP). I/O controller hub (Primary/Secondary IDE, PCI, Serial and Parallel ports, PS/2, USB). BIOS ROM. CMOS battery. Real time clock. SMPS connector.
- 5. COMPUTER NETWORKS** — Concept of network (LAN, MAN, WAN). Transmission modes (Simplex, Half-duplex, Full-duplex). Transmission media (Co-axial, Twisted pair UTP/STP, Optical Fiber). Topologies (Bus, Star, Ring, Mesh). Architectures (Peer-to-Peer, Client-Server). Connectivity devices (MODEM, Repeater, Router, Bridge, Gateway). OSI reference model — all 7 layers. Standards and Protocols: TCP/IP, NetBEUI, IPX/SPX, IEEE Standards.
- 6. INTERNET TECHNOLOGY** — Basics of Intranet and Internet. Services: Email, Chatting, Bulletin Boards, Video conferencing, FTP, Telnet. WWW: Web-site access and information search, Browsers and search engines. Connectivity: ISP, Internet accounts — Dial-up, Leased line, ISDN, VSAT.
- 7. COMPUTER LANGUAGES** — Analogy with natural languages. Characteristics of a good language. Classification: Machine, Assembly, High-level.
- 8. COMPUTER SOFTWARE** — Relationship between hardware and software. Classification. System software (OS, Compiler & Interpreter). Application software (MS Office, PageMaker, Photoshop). **MS Office in detail:** MS WORD (components, creating/editing, formatting, symbols/picture/WordArt, tabs/borders/shading, header/footer, footnote/endnote, shapes, text box/callouts/captions, tables, autocorrect, spelling & grammar, DropCap, multiple columns, Mail Merge); MS EXCEL (features, components, worksheets, formatting/editing, formulas, types of functions, sharing/managing data, charts — components, types, 3D); MS POWERPOINT (components, views, design templates, new presentation, title slide, formatting, transition effects, action buttons, animation); MS ACCESS (database planning, creating, adding records, querying, forms, reports).
- 9. COMPUTER VIRUSES** — Introduction. Types: Command processor infection, Boot sector infection, Executable file infection, File-specific infection, Memory resident infection, Macro viruses. Protecting the PC. Recognizing an infection. Dealing with an infection.

Paper-II

- 1. OPERATING SYSTEM** — Definition, Functions, Evolution (Sequential, Batch, Multiprogramming, Multiprocessing, Real time, Multitasking). Process concept. CPU scheduling — FCFS, SJF, Priority, Round Robin, Multiprocessor scheduling. Memory management. I/O Systems and Mass Storage Structure. **DOS:** Features & characteristics of MS DOS, Booting, system files, BIOS, Files and directory, Internal commands, External commands, meaning of BAT/EXE/COM files. **Windows OS:** Arranging/Moving/Resizing/Opening/Closing windows, Folder/File management (search, copy, delete, rename, Restore), Accessories (Notepad, WordPad, Paint), Overview of MS Windows versions.

2. **UNIX OPERATING SYSTEM** — History and development. Features, Structure (Kernel, Shell, Application Utilities). Hardware requirements for installation. Essential UNIX commands. Startup & Shutdown, Booting procedure stages, Login process, password concept, `who`, `who am i`, `tty`, `date`, `cal`, system shutdown. File concept: file types, hierarchical directory structure, file system structure. File creating/displaying/concatenating/copying. Creating and changing directories, removing files/directories — `cd`, `cp`, `md`, `rm`, `mkdir`, `cat`. File processing commands — `wc`, `head`, `tail`, `cut`, `paste`, `join`, `split`, `sort`, `grep`, `egrep`, `tr`, `comm`, `cmp`, `diff`, `more`, `less`. File formatting and printing — `pr` with options, `lp`. Online help — `man`, `help`. Mathematical commands — `bc`, `expr`, `factor`, `units`. Communication commands — `write`, `mail`, `wall`.
3. **DATABASE MANAGEMENT SYSTEM** — Basic concepts of Data and Information: Database, DBMS, Database Administrator, Functions of DBMS. Relational Databases: relational system, relational model, optimization, base tables and views, the SQL language. Relational Data Objects: domains and relations, kinds of relations, relations and predicates, relational database. The Entity/Relationship Model: overall approach, overview, E/R diagrams, database with the E/R model. SQL Language: data definition, data manipulation, retrieval operation, update operation, table expressions, conditional expressions, scalar expressions, embedded SQL. Database security: authentication, authorization, access control, enforcement.
4. **SYSTEM ANALYSIS AND DESIGN** — System concept: definition, characteristics, elements. Types: Physical, Abstract, Open, Closed. Information system. SDLC: different phases/stages, Role of the System Analyst, Factors affecting failure in SDLC.
5. **PROGRAMMING IN 'C' LANGUAGE** — History and features of C, Algorithms, Flowcharts, Structured programming concepts. Operations & Expressions: character set, operators (arithmetic, relational, logical, assignment), variables, constants, data types, expressions, data type conversion, keywords, hierarchy of operators. Input & Output: program structure, type declaration, `printf`, `scanf`, `getchar`, `putchar`, `getch`, `putch`, conversion specifiers, library (Math) functions. Control statements: `goto`, `if`, `if-else`, nested `if`, `switch-case`. Loops: `for`, `while`, `do-while`, `break`, `continue`. (Continues with arrays, functions, pointers, structures and file handling — see source PDF pages 9–11.)

5. THE OVERLAP — YOUR SINGLE BIGGEST ADVANTAGE

You sit **three** electives, but they are **not** three separate syllabi. Study the shared core once and it pays into all three papers.

Topic	CS Degree	CS Diploma	Comp. Forensic
Number systems, 1's/2's complement, Boolean algebra, logic gates, K-maps, flip-flops, counters	P-I §3	P-I §1.3	TP-I §6
Computer Organization & Architecture (addressing modes, memory hierarchy, cache, DMA, I/O)	P-I §4	P-I §3,4	TP-I §9
Operating Systems (scheduling, memory mgmt, deadlock, file systems)	P-II §1	P-II §1	TP-I §8
UNIX / Linux commands & file system	—	P-II §2	TP-I §5, §8
Computer Networks (OSI, TCP/IP, topologies, media, devices, routing, IP)	P-II §4	P-I §5,6	TP-I §7
DBMS + SQL + E-R + normalization	P-II §3	P-II §3	TP-II §5
Software Engineering / SDLC	P-II §2	P-II §4	TP-II §3
Data Structures (array, stack, queue, linked list, tree, graph, hashing, B-trees)	P-I §2	—	TP-II §6
Programming in C / C++ / OOP	P-I §1	P-II §5	TP-II §4

Web technologies (HTML, scripting)	P-II §5	—	TP-II §4
Mobile computing / applications	P-I §6	—	TP-II §7
Computer hardware, motherboard, storage devices	—	P-I §2,3,4	TP-I §4
Viruses, malware, security threats	—	P-I §9	TP-I §8, TP-II §1

Unique-to-one-paper topics (cannot be shared — budget separate time):

- **CS Degree only:** Compilers & System Software (P-I §5), AI (P-II §6), Machine Learning (P-II §7), Algorithm analysis / NP-completeness (P-I §2).
- **CS Diploma only:** MS Office in depth (Word/Excel/PowerPoint/Access) (P-I §8), MS-DOS commands (P-II §1).
- **Computer Forensic only:** Forensic Science fundamentals (TP-I §1,2,3), Windows/Linux/Mac artifacts (TP-I §5), Cyber crime & first responder (TP-II §1), Browser & email forensics (TP-II §2), Digital evidence handling & hashing (TP-II §8).

Strategy consequence: roughly **60%** of your total technical load is shared core.

Master the shared core first — it is the highest-return study you can do.